

Lab 05:

Question#01:

CODE:

```
#include <iostream>
using namespace std;

class BaseClass {
    private:
        int privateInt;
    protected:
        int protectedInt;
    public:
        int publicInt;

        void setPrivateInt(int value) {
            privateInt = value;
        }

        void setProtectedInt(int value) {
            protectedInt = value;
        }

        void setPublicInt(int value) {
            publicInt = value;
        }

        int getPrivateInt() {
            return privateInt;
        }

        int getProtectedInt() {
            return protectedInt;
        }

        int getPublicInt() {
            return publicInt;
        }
};
```

Object Oriented Programming

```
class PublicClass : public BaseClass {
public:
    void display() {
        protectedInt = 20; // Accessible
        publicInt = 30;    // Accessible
        // privateInt = 40; // Not Accessible

        cout << "ProtectedInt = " << protectedInt << endl;
        cout << "PublicInt = " << publicInt << endl;
    }
};

class ProtectedClass : protected BaseClass {
public:
    void display() {
        protectedInt = 40; // Accessible
        publicInt = 50;    // Accessible
        // protectedInt = 60; // Not Accessible

        cout << "ProtectedInt = " << protectedInt << endl;
        cout << "PublicInt = " << publicInt << endl;
    }
};

class PrivateClass : private BaseClass {
public:
    void display() {
        protectedInt = 60; // Accessible
        publicInt = 70;    // Accessible
        // protectedInt = 80; // Not Accessible

        cout << "ProtectedInt = " << protectedInt << endl;
        cout << "PublicInt = " << publicInt << endl;
    }
};

int main() {
    PublicClass obj1;

    obj1.setPrivateInt(10);
    obj1.setProtectedInt(20);
    obj1.setPublicInt(30);
}
```

Object Oriented Programming

```
    cout << "Accessing PrivateInt using getter: " << obj1.getPrivateInt() <<
endl; // Accessible using getters

    obj1.display();

    cout << "-----" << endl;

    ProtectedClass obj2;
    obj2.display(); // obj2 setPublicInt is not accessible outside the class as
it is protected.

    cout << "-----" << endl;

    PrivateClass obj3;
    obj3.display(); // obj3 setPublicInt is not accessible as it is now private.


    return 0;
}
```

OUTPUT:

```
Accessing PrivateInt using getter: 10
ProtectedInt = 20
PublicInt = 30
-----
ProtectedInt = 40
PublicInt = 50
-----
ProtectedInt = 60
PublicInt = 70
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 05> |
```

Question#02:

CODE:

```
#include <iostream>
using namespace std;

class Teacher {
protected:
    string Name;
    int Age;
    string Institute;

public:
    // Setters
    void setName(string name) {
        Name = name;
    }

    void setAge(int age) {
        Age = age;
    }

    void setInstitute(string institute) {
        Institute = institute;
    }

    // Getters
    string getName() {
        return Name;
    }

    int getAge() {
        return Age;
    }

    string getInstitute() {
        return Institute;
    }
};

class HumanitiesTeacher : public Teacher {
private:
    string Department = "Humanities";
    string CourseName;
```

Object Oriented Programming

```
    string Designation;

public:
    void setCourseName(string course) {
        CourseName = course;
    }

    void setDesignation(string designation) {
        Designation = designation;
    }

    void display() {
        cout << "\n--- Humanities Teacher ---\n";
        cout << "Name: " << Name << endl;
        cout << "Age: " << Age << endl;
        cout << "Institute: " << Institute << endl;
        cout << "Department: " << Department << endl;
        cout << "Course Name: " << CourseName << endl;
        cout << "Designation: " << Designation << endl;
    }
};

class ScienceTeacher : public Teacher {
private:
    string Department = "Science";
    string CourseName;
    string Designation;

public:
    void setCourseName(string course) {
        CourseName = course;
    }

    void setDesignation(string designation) {
        Designation = designation;
    }

    void display() {
        cout << "\n--- Science Teacher ---\n";
        cout << "Name: " << Name << endl;
        cout << "Age: " << Age << endl;
        cout << "Institute: " << Institute << endl;
        cout << "Department: " << Department << endl;
```

Object Oriented Programming

```
        cout << "Course Name: " << CourseName << endl;
        cout << "Designation: " << Designation << endl;
    }
};

class MathsTeacher : public Teacher {
private:
    string Department = "Maths";
    string CourseName;
    string Designation;

public:
    void setCourseName(string course) {
        CourseName = course;
    }

    void setDesignation(string designation) {
        Designation = designation;
    }

    void display() {
        cout << "\n--- Maths Teacher ---\n";
        cout << "Name: " << Name << endl;
        cout << "Age: " << Age << endl;
        cout << "Institute: " << Institute << endl;
        cout << "Department: " << Department << endl;
        cout << "Course Name: " << CourseName << endl;
        cout << "Designation: " << Designation << endl;
    }
};

int main() {

    HumanitiesTeacher h;
    ScienceTeacher s;
    MathsTeacher m;

    string name, institute, course, designation;
    int age;

    cout << "Enter Humanities Teacher Details:\n";
```

Object Oriented Programming

```
cout << "Name: ";
cin >> name;
cout << "Age: ";
cin >> age;
cout << "Institute: ";
cin >> institute;
cout << "Course Name: ";
cin >> course;
cout << "Designation: ";
cin >> designation;

h.setName(name);
h.setAge(age);
h.setInstitute(institute);
h.setCourseName(course);
h.setDesignation(designation);

cout << "\nEnter Science Teacher Details:\n";
cout << "Name: ";
cin >> name;
cout << "Age: ";
cin >> age;
cout << "Institute: ";
cin >> institute;
cout << "Course Name: ";
cin >> course;
cout << "Designation: ";
cin >> designation;

s.setName(name);
s.setAge(age);
s.setInstitute(institute);
s.setCourseName(course);
s.setDesignation(designation);

cout << "\nEnter Maths Teacher Details:\n";
cout << "Name: ";
cin >> name;
cout << "Age: ";
cin >> age;
cout << "Institute: ";
cin >> institute;
cout << "Course Name: ";
cin >> course;
```

Object Oriented Programming

```
    cout << "Designation: ";
    cin >> designation;

    m.setName(name);
    m.setAge(age);
    m.setInstitute(institute);
    m.setCourseName(course);
    m.setDesignation(designation);

    h.display();
    cout << "-----" << endl;
    s.display();
    cout << "-----" << endl;
    m.display();

    return 0;
}
```

OUTPUT:

Object Oriented Programming

```
Enter Humanities Teacher Details:
Name: Zainab
Age: 18
Institute: KU
Course Name: H-240
Designation: Teacher
```

```
Enter Science Teacher Details:
Name: Samia
Age: 24
Institute: TCS
Course Name: S-650
Designation: Teacher
```

```
Enter Maths Teacher Details:
Name: Shazaib
Age: 30
Institute: NED
Course Name: M-280
Designation: Teacher
```

```
--- Humanities Teacher ---
Name: Zainab
Age: 18
Institute: KU
Department: Humanities
Course Name: H-240
Designation: Teacher
-----
```

```
--- Science Teacher ---
Name: Samia
Age: 24
Institute: TCS
Department: Science
Course Name: S-650
Designation: Teacher
-----
```

```
--- Maths Teacher ---
Name: Shazaib
Age: 30
Institute: NED
Department: Maths
Course Name: M-280
Designation: Teacher
```

```
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 05> █
```

Question#03:

CODE:

```
#include <iostream>
using namespace std;

class Weapons {
public:
    void WeaponsDescription() {
```

Object Oriented Programming

```
        cout << "Weapons are tools which are used to cause damage or destruction.  
Sometimes they are used in self=defense." << endl;  
    }  
};  
  
class HotWeapons : public Weapons {  
public:  
    void HotWeaponsDescription() {  
        cout << "Hot weapons use gunpowder, heat, or explosive materials to  
function." << endl;  
    }  
};  
  
class Bombs : public HotWeapons {  
public:  
    void BombsDescription() {  
        cout << "Bombs are explosive weapons that blow up targets. i.e Grenades,  
Mines etc." << endl;  
    }  
};  
  
class NuclearBombs : public Bombs {  
public:  
    void NuclearBombsDescription() {  
        cout << "Nuclear bombs cause massive explosions using nuclear fission and  
fusion." << endl;  
    }  
};  
  
int main() {  
  
    NuclearBombs nb;  
  
    // Because of multilevel inheritance,  
    // NuclearBombs can access all parent methods  
  
    nb.WeaponsDescription();  
    nb.HotWeaponsDescription();  
    nb.BombsDescription();  
    nb.NuclearBombsDescription();  
  
    return 0;  
}
```

OUTPUT:

```
}; if ($?) { .\CT-25276_Q3 }  
Weapons are tools which are used to cause damage or destruction. Sometimes they are used in self-defense.  
  
Hot weapons use gunpowder, heat, or explosive materials to function.  
  
Bombs are explosive weapons that blow up targets. i.e Grenades, Mines etc.  
  
Nuclear bombs cause massive explosions using nuclear fission and fusion.  
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 05> █
```

Question#04:

CODE:

```
#include <iostream>  
using namespace std;  
  
// ===== Base Class =====  
class Item {  
protected:  
    string Name;  
    int Quantity;  
  
public:  
    void setItem(string name, int qty) {  
        Name = name;  
        Quantity = qty;  
    }  
  
    string getName() {  
        return Name;  
    }  
  
    int getQuantity() {  
        return Quantity;  
    }  
};  
  
class BakedGoods : public Item {  
protected:
```

Object Oriented Programming

```
        double discount = 0.10;

public:
    double applyDiscount(double total) {
        return total - (total * discount);
    }
};

class Cakes : public BakedGoods {
private:
    double price = 600;

public:
    double calculateBill() {
        double total = price * Quantity;
        return applyDiscount(total);
    }
};

class Bread : public BakedGoods {
private:
    double price = 200;

public:
    double calculateBill() {
        double total = price * Quantity;
        return applyDiscount(total);
    }
};

class Drinks : public Item {
private:
    double price = 100;
    double discount = 0.05;

public:
    double calculateBill() {
        double total = price * Quantity;
        return total - (total * discount);
    }
};
```

Object Oriented Programming

```
int main() {

    int choice, qty;
    string name;

    while (true) {
        cout << "Select Item:\n";
        cout << "1. Cake (600, 10% discount)\n";
        cout << "2. Bread (200, 10% discount)\n";
        cout << "3. Drink (100, 5% discount)\n";
        cout << "4. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        if (choice == 4) {
            cout << "Exiting..." << endl;
            break;
        }

        if (choice < 1 || choice > 4) {
            cout << "Invalid choice. " << endl;
            continue;
        }

        cout << "Enter quantity: ";
        cin >> qty;

        if (choice == 1) {
            Cakes c;
            c.setItem("Cake", qty);
            cout << "Total Bill: " << c.calculateBill() << "\n" << endl;
        }
        else if (choice == 2) {
            Bread b;
            b.setItem("Bread", qty);
            cout << "Total Bill: " << b.calculateBill() << "\n" << endl;
        }
        else if (choice == 3) {
            Drinks d;
            d.setItem("Drink", qty);
            cout << "Total Bill: " << d.calculateBill() << "\n" << endl;
        }
    }
}
```

Object Oriented Programming

```
    return 0;  
}
```

OUTPUT:

```
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 05> cd "d:\NED U  
cticals\Lab 05\" ; if ($?) { g++ CT-25276_Q4.cpp -o CT-25276_Q4 } ; if ($?) { .\CT-25276_Q4 }  
Select Item:  
1. Cake (600, 10% discount)  
2. Bread (200, 10% discount)  
3. Drink (100, 5% discount)  
4. Exit  
Enter choice: -2  
Invalid choice.  
Select Item:  
1. Cake (600, 10% discount)  
2. Bread (200, 10% discount)  
3. Drink (100, 5% discount)  
4. Exit  
Enter choice: 1  
Enter quantity: 5  
Total Bill: 2700  
  
Select Item:  
1. Cake (600, 10% discount)  
2. Bread (200, 10% discount)  
3. Drink (100, 5% discount)  
4. Exit  
Enter choice: 2  
Enter quantity: 5  
Total Bill: 900  
  
Select Item:  
1. Cake (600, 10% discount)  
2. Bread (200, 10% discount)  
3. Drink (100, 5% discount)  
4. Exit  
Enter choice: 3  
Enter quantity: 4  
Total Bill: 380
```

Object Oriented Programming

```
Select Item:
1. Cake (600, 10% discount)
2. Bread (200, 10% discount)
3. Drink (100, 5% discount)
4. Exit
Enter choice: 6
Invalid choice.
Select Item:
1. Cake (600, 10% discount)
2. Bread (200, 10% discount)
3. Drink (100, 5% discount)
4. Exit
Enter choice: 4
Exiting...
PS D:\NED University\2nd Semester\Object Oriented Programming (OOPs)\Practicals\Lab 05>
```